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Desigualdad de Chebyshev

Proposición 1 (Desigualdad de Chebyshev). *Sea X una v.a. y $\varphi\colon \mathbb{R} \longrightarrow \mathbb{R}$ medible y no negativa, definimos $\forall A \in \mathcal{B}(\mathbb{R}) : i_A := \inf_{t\in A} \varphi(t)$*

$$\implies i_A \cdot \mathbb{P}\left(X \in A\right) \leqslant \mathbb{E}\left[\mathbb{1}_{\{X\in A\}} \cdot \varphi(X)\right].$$

***Demostración*:** Sea $A \in \mathcal{B}(\mathbb{R})$, entonces $\forall \omega \in \Omega : X(\omega) \in A : i_A \leqslant \varphi(X(\omega))$

$$\implies i_A \cdot \mathbb{1}_{\{X\in A\}}(\omega) \leqslant \mathbb{1}_{\{X\in A\}}(\omega) \cdot \varphi(X(\omega))$$

Integrando, obtenemos que $i_A \cdot \mathbb{P}\left(X \in A\right) \leqslant \mathbb{E}\left[i_A \cdot \mathbb{1}_{\{X\in A\}}\right] \leqslant \mathbb{E}\left[\mathbb{1}_{\{X\in A\}} \cdot \varphi(X)\right]$. ■

Referenciado en

- `teo-convergencia-martingalas`
- `convergencia-lp-imp-probabilidad`
- `prop-borel-cantelli-iii`
- `desigualdad-markov`
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